

Lesson 1 Definitions, Theorems, and Proofs

1. Determine if each of the following is a statement. If it is, is it true or false?
 - a) $2 + 3$
 - b) $x > y$
 - c) An integer is a real number.
 - d) A real number is an integer.
 - e) Goldbach's conjecture is true.
 - f) What is the four color theorem?
 - g) Every positive number
 - h) The sum of two odd numbers is odd.
2. Prove that:
 - a) -212 is even
 - b) -101 is odd
3. Answer Question 2, using the following definitions:
 - An integer x is **even** if $x/2$ is an integer.
 - An integer x is **odd** if $x = 2n - 1$ for some integer n .
4. Prove or disprove:
 - a) $0|0$
 - b) $0|2$
 - c) $2|0$
 - d) $2|2$
 - e) $(-2)|2$
 - f) $2|(-4)$
5. Prove or disprove:
 - a) 1 is prime
 - b) 2 is prime
 - c) 91 prime
6. Complete the following definition:
 - Two primes p and q are **twins** if _____
and give the first five pairs of twin primes.
7. Complete the following definition:
 - Two integers have the **same parity** if _____
Use your definition to prove or disprove that 1 and -1 have the same parity.
8. Consider the following definition:
 - A positive integer n is **perfect**, **deficient**, or **abundant**, respectively, if the sum of its positive divisors is equal to, less than, or greater than, $2n$.
 - a) Determine if 18 is a perfect, abundant, or deficient number. Give a proof.
 - b) What is the *smallest* perfect number? Prove it.

9. Let x be a real number.

- The **floor** of x , $\lfloor x \rfloor$, is the largest integer less than or equal to x .
- The **ceiling** of x , $\lceil x \rceil$, is the least integer greater than or equal to x .
- The **integral part** of x is $[x] = \begin{cases} \lfloor x \rfloor, & x \geq 0 \\ \lceil x \rceil, & x < 0 \end{cases}$
- The **absolute value** of a real number x is $|x| = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}$

Find the floor, ceiling, integral part, and absolute value of -0.5 , 0 , and 0.5 .

10. An integer n is a **linear combination** of integers a and b if $n = ax + by$, where x and y are integers. Prove that 1 is a linear combination of 3 and 5 .

11. Let a and b be integers.

- An integer c is a **common divisor** of a and b if $c|a$ and $c|b$, and the **greatest common divisor** of a and b is denoted $\gcd(a, b)$.
 - An integer m is a **common multiple** of a and b if $a|m$ and $b|m$. The **least common multiple** of a and b , denoted $\text{lcm}(a, b)$, is the smallest positive common multiple of a and b .
 - Two integers a and b are **relatively prime** if $\gcd(a, b) = 1$
- a) Find $\gcd(20, 50)$.
 - b) Find $\text{lcm}(20, 50)$.
 - c) Prove or disprove: 9 and 16 are relatively prime

12. A conjecture is a statement which is probably true but has not been proved or disproved, a famous example is Goldbach's conjecture:

Every even integer greater than 2 is the sum of two primes.

Show that Goldbach's conjecture is true for all even integers greater than 2 up to and including 20 .